

MUHAMMAD SARMAJ SAJJAD

Computer Science Graduate

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PROFESSIONAL SUMMARY

Computer Science graduate (GPA: 3.74) with hands-on experience in software development, technical research, web applications, APIs, databases, and cloud fundamentals. Strong analytical, communication, problem-solving, and research skills with experience collaborating across teams and explaining technical concepts clearly. Interested in technology-driven business initiatives, market research, client engagement, and growth opportunities.

BUSINESS & TECHNICAL SKILLS

Business: Market Research, Competitive Analysis, Lead Research, Client Outreach Support, Technical Research

Communication: Client Communication, Presentation Support, Team Collaboration, Documentation

Research & Analysis: Data Analysis, Problem Solving, Requirements Understanding, Technology Evaluation

Technology: Software Development, Web Applications, Full-Stack Development, REST APIs, Database Systems, Cloud Fundamentals, AI/ML, RAG, Multi-Agent Systems, Deep Learning, CV, Transfer Learning.

Programming: Python, JavaScript, TypeScript, SQL, C++, HTML, CSS

Frameworks & Tools: Flask, FastAPI, Django, React, Next.js, Node.js, Express.js, LangChain, LangGraph, Git, Docker, OpenCV, PyTorch, scikit-learn

Databases: MySQL, PostgreSQL, MongoDB, SQLite, Pinecone, Relational Database Design, SQL Queries

Office & Collaboration: MS Office, Digital Collaboration Tools

PROFESSIONAL EXPERIENCE

AI/ML Intern — Texense Sep 2025 – Dec 2025 | Remote

- Worked on software solutions and computer-vision applications, gaining practical exposure to technology-driven product development.
- Researched and evaluated technical approaches for improving application performance, reliability, and real-time processing.
- Collaborated with cross-functional team members to troubleshoot issues and refine project requirements and solutions.
- Developed FastAPI-based model-serving endpoints and integrated trained systems into real-time applications.
- Worked with Python, PyTorch, OpenCV, and optimization techniques including quantization and mixed-precision inference.

Freelance Web Developer — Self-Employed Jun 2024 – Sep 2025 (1 yr 3 mos) | Remote

- Designed, built, and maintained responsive websites and web applications for independent clients alongside academic commitments.
- Managed end-to-end freelance engagements including requirements gathering, client communication, development, testing, and delivery.
- Built front-end interfaces and back-end functionality using modern web technologies and integrated databases for dynamic content.
- Handled project timelines and client feedback independently, strengthening client communication and self-management skills.

Teaching Assistant — University of Management and Technology Mar 2024 – Aug 2025 | Lahore

- Supported 100+ students in Machine Learning, Data Structures & Algorithms, and OOP labs.
- Communicated technical concepts clearly and assisted students in understanding project requirements and implementation approaches.
- Reviewed projects and provided feedback to identify issues and improve solutions.
- Guided students through problem-solving, debugging, and technical decision-making.

SELECTED PROJECTS

AI Real Estate Agent Team

- Built a multi-agent property search and analysis platform using the Agno framework and Firecrawl's Extract API, with dedicated agents for property search, market analysis, and valuation.
- Aggregated and analyzed listings across Zillow, Realtor.com, Trulia, and Homes.com, producing market insights, price-trend analysis, and investment recommendations via an interactive Streamlit UI.

AI Fraud Investigation Agent

- Built an autonomous investigation agent that cross-references childcare licensing records with property, business-registration, and geolocation data to flag potential licensing anomalies for review.
- Integrated Illinois DCFS licensing data, Cook County property records, Google Places/Street View, and the Illinois Secretary of State's business registry, applying occupancy-code calculations and pattern detection across providers.

Document Intelligence System

- Built an offline, multilingual document-intelligence pipeline supporting 200+ languages, extracting named entities, relationships, topics, sentiment, and summaries from PDFs, DOCX, images, and chat exports.
- Implemented a dual-language RAG architecture (NLLB-200 translation, GLiNER NER, BERTopic, Qwen2.5, BGE-M3 embeddings with Qdrant, and a Neo4j knowledge graph) enabling multilingual Q&A with original-language citations.

Sehat Guftagu – AI Clinical Interview Platform

- Built a bilingual (Urdu/English) AI clinical-interview platform coordinating 5 specialized LangGraph agents to conduct medical interviews, generate SOAP reports, and route cases to a doctor review portal.
- Implemented voice and text interview modes, automatic emergency red-flag triage, and a medical RAG knowledge base using Next.js, PostgreSQL/Prisma, Groq LLaMA 3.3, Pinecone, and ElevenLabs.

Techy Electronics – MERN Stack E-Commerce Platform

- Built and deployed a full-stack e-commerce app (React, Node.js, Express, MongoDB) with product search, cart, simulated checkout, order tracking, and JWT authentication.
- Added Pinecone-based product recommendations, Swagger API documentation, Jest tests, and Docker/GitHub Actions setup.

College ERP System

- Built a Django ERP with Admin, Staff, and Student portals and role-based access, covering attendance, results, leave approvals, feedback, and dashboards.

XAI-Based Breast Cancer Detection

- Developed an explainable system for multi-class mammogram classification using multiple neural-network architectures, ensemble learning, SHAP, and Grad-CAM.

AI Image Classification System

- Developed an image classification system with real-time prediction through a Flask web application. Technologies: Python, PyTorch, CNN, Flask, OpenCV.

Food Restaurant Website

- Built the front-end for a restaurant website using HTML, CSS, and JavaScript, with light PHP for basic backend functionality.

EDUCATION

Bachelor of Computer Science — University of Management and Technology, Lahore *Oct 2022 – Jul 2026 | GPA: 3.74*

Intermediate — Punjab Group of Colleges *2020 – 2022*

Matriculation — Govt. Islah-e-Moashra High School, Lahore *2020*

CERTIFICATIONS & ACHIEVEMENTS

- AWS Academy Graduate – Cloud Foundations – Training Badge | Amazon Web Services | Nov 2025
- ICCET | Superior University | Jan 2026
- Rector Merit Award Winner — UMT | May 2025 | 4.00/4.00 GPA
- Dean's Merit Award — UMT | Jan 2025
- Certificate of English Immersion — University of Management and Technology (UMT)
- Certificate of 21st Century Skills — University of Management and Technology (UMT)

LANGUAGES

English | Urdu